Proposed System Architecture (The "Why")

Before jumping into the diagram, Slide 3 sets the stage. If the panel asks for the core innovation of your Type 2 project, you hit them with these three pillars:

1. **Hybrid Deep Learning Model:** We aren't just looking at static data. We use a dual-branch network (CNNs + LSTMs) to understand both the physical "shape" of the data (Spatial) and the sequence of movements over time (Temporal).
2. **Attention-Enhanced Fusion:** This is the project's secret weapon. Sensors fail or get noisy. Our Attention Mechanism acts as a smart manager, dynamically prioritizing the most reliable sensor for a specific task (e.g., trusting the wearable for a wrist movement, but trusting the ambient sensor if the wearable is taken off).
3. **Synchronized Preprocessing Pipeline:** Our hardware (Bazel board, Wi-Fi, wearables) collects data at wildly different speeds. This pipeline forces everything into a perfectly timed, standardized matrix so the neural network can actually read it.

Slide 3.1: Workflow Diagram (The "How")

When you switch to Slide 3.1, point to the far left of your diagram and physically walk the panel through the data's journey to the final output.

1. Data Input Layer (The Raw Sensors)

This is where we capture the physical world without using privacy-invading cameras.

- **Ambient Sensors:** (e.g., Motion detectors, door contact sensors). They provide high-level context ("someone is in the kitchen") but suffer from *Identity Ambiguity*—they don't know who triggered them or what exact action is happening.
- **Wearable Sensor:** (e.g., Wrist IMU/Accelerometer). This solves the identity problem. It tags *who* is moving and captures fine motor skills (like chopping vegetables). The flaw? Users forget to wear them.
- **Wi-Fi CSI (Channel State Information):** The "invisible radar." It measures how Wi-Fi waves bounce off humans in the room. It offers device-free tracking of breathing and movements, replacing the need for cameras.

2. Pre Processing Layer (The Cleaner)

Raw data is messy. If we feed it directly to the AI, it will fail. This layer cleans and standardizes the streams.

- **Data Normalization:** Sensors speak at different "volumes." Wi-Fi outputs massive complex numbers, while a door sensor just outputs a 0 or 1. Normalization scales all values to a common range (like 0 to 1) so the neural network treats them equally.
- **Segmentation:** Continuous data streams are chopped into overlapping chunks called "sliding windows" (e.g., 2-second blocks). This gives the AI a perfectly sized frame to analyze without missing an action that happens between cuts.
- **Temporal Alignment:** Our hardest engineering challenge. Wi-Fi records hundreds of times a second; ambient sensors record only when triggered. We use math (like forward-padding) to sync the timestamps perfectly.
- **The Handoff:** This layer outputs a clean, synchronized 2D matrix of data for that 2-second window.

3. Feature Extraction Layer (The Detective)

This layer takes the clean matrix and looks for hidden mathematical patterns (Feature Vectors). It operates in two parallel branches:

- **Spatial Features (CNN):** Convolutional Neural Networks look at the data *at a frozen moment in time*. It maps out the physical correlations—for example, the relationship between the Wi-Fi distortion and the wearable's exact angle. It asks: *"What is the physical state right now?"*
- **Temporal Features (LSTM):** Long Short-Term Memory networks look at the *sequence*. Human activity isn't static. The LSTM tracks the rhythm and speed across the sliding window to capture the motion of a fall versus the motion of sitting down. It asks: *"How is this state changing over time?"*
- **The Handoff:** The CNN and LSTM output their extracted patterns (Feature Vectors) and send them to the Fusion Module.

4. Fusion Module (The Judge)

This is where the magic happens and where the multi-resident ambiguity problem is officially solved.

- **Feature-Level Fusion:** We stitch the raw spatial and temporal patterns together *before* the model makes a guess. This helps the system find deep cross-correlations (e.g., pairing an ambient fridge trigger directly with a Wi-Fi wave shift).
- **Decision-Level Fusion:** Different branches of the network calculate their own independent probabilities (e.g., the Wi-Fi data thinks it's a fall, the wearable thinks it's a fall). We combine these "votes" for maximum confidence, ensuring a glitch in one sensor doesn't ruin the whole system.

- **Attention Mechanism:** The system dynamically calculates weights. If a user takes off their wearable to shower, the wearable data goes flat. The Attention layer instantly realizes this, lowers the wearable's weight to near zero, and shifts 100% of its focus to the Wi-Fi CSI and ambient bathroom sensors.

5. Classification Layer (The Verdict)

All that dense math needs to be translated into an actionable smart home trigger.

- **Activity Classification:** Using a Fully Connected Dense network and a Softmax activation function, this layer takes the fused, attention-weighted data and calculates a final percentage for predefined classes.
- **The Output:** It spits out a clear, real-time label (e.g., 95% confidence of "Cooking", or 98% confidence of "Fall Detected"). This final label is what the Bazel board will use in Phase 2 to physically trigger an alarm or send a medical alert.

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1. **Privacy:** We are replacing invasive cameras with Wi-Fi CSI and ambient data.
2. **Robustness:** If one sensor fails (like forgetting a wearable), the Attention Mechanism ensures the system still works.
3. **Context + Identity:** Ambient gives the location context; wearables and Wi-Fi give the specific physical action and user identity.



How this fits into “Temporal Alignment”

Temporal alignment = **three operations together**

Sensor type	Operation
Very fast (WiFi)	Downsampling / averaging
Medium speed (wearables)	Interpolation
Slow / event-based (ambient)	Forward fill / padding